Some Useful Concepts from Information Theory

COMP9418 — Advanced Topics in Statistical Machine Learning

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Last Update: Sunday 12th August, 2018 at 22:38

Acknowledgments

- [Cover & Thomas, EoIT, 2012] Elements of Information Theory. Thomas M. Cover and Joy A. Thomas. John Wiley & Sons, 2012.
- [Mackay, ITILA, 2003] Information Theory, Inference, and Learning Algorithms . David J. C. Mackay. Cambridge University Press. 2003.
- [Bishop, PRML, 2006] Pattern Recognition and Machine Learning, Christopher Bishop, 2006
- [Murphy, MLaPP, 2012] Machine Learning: A Probabilistic Perspective, Kevin P. Murphy, 2012

1. Information Content & Entropy
    - Entropy of a Random Variable
    - Some Useful Properties
    - Examples
    - Maximum Entropy
2. Joint Entropy and Conditional Entropy
3. Relative Entropy (KL Divergence) and Mutual Information
    - Mutual Information for Feature Selection in Machine Learning
    - Chain Rule for Mutual Information
4. Some Important Inequalities
    - Jensen's Inequality
    - Gibbs' Inequality
    - Information Cannot Hurt
    - Data Processing Inequality

Information Content

Information as:

- Amount of unexpected data
- message that its uncertain to receivers
  - If we are told that a very likely event has happened vs an unlikely event

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- Information content of an outcome must depend on its probability
- Information content of a random variable must depend on its probability distribution

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Entropy (or information content) of an outcome x:

$$h(x) = \log_2 \frac{1}{p(x)}$$

- Choice of logarithm basis is arbitrary
- If we use $\log_2$ we measure information in *bits*

Entropy of a Random Variable

Let X be a discrete r.v. with values drawn from alphabet $\mathcal{X}$ from a total number of states $|\mathcal{X}|$.

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If we want to transmit the value of X, the average amount of information is given by:

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- The expectation of the entropy of each outcome wrt $p(x)$

We call this the entropy of the r.v. X (Note dependency on distribution)

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Note that we can write:

$$H(X) = -\sum_x p(x) \log_2 p(x)$$

and define: $0 \log 0 \equiv 0$, as $\lim_{p \to 0} p \log p = 0$.

- Non-negativity:

$$0 \le p(x) \le 1 \rightarrow \log \frac{1}{p(x)} \ge 0$$

$$\sum_x p(x) \log \frac{1}{p(x)} \ge 0$$

$$H(X) \ge 0$$

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- Change of base:

$$H_b(X) = -\sum_x p(x) \log_b p(x)$$

$$= \sum_x p(x) \log_a p(x) \log_b a$$

$$H_b(X) = \log_b a \; H_a(X)$$

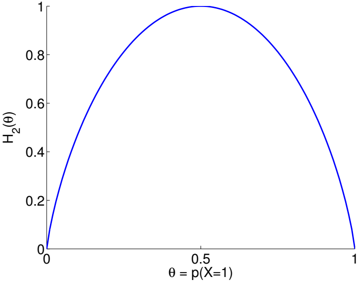
  - If we use natural logarithm the units are call *nats*

Entropy of a Random Variable

Example 1 — Bernoulli Distribution

Let $X \in \{0, 1\}$ with $X \sim \text{Bern}(X|\theta)$ and $\theta = p(X = 1)$:

$$H(X) = -\theta \log \theta - (1 - \theta) \log(1 - \theta)$$ Why?



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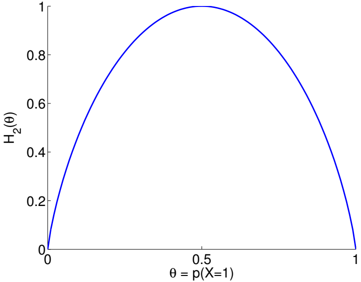

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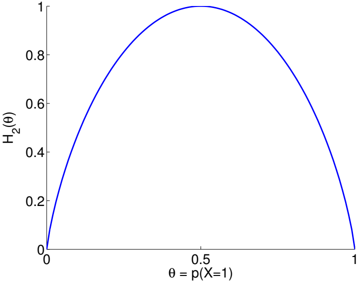

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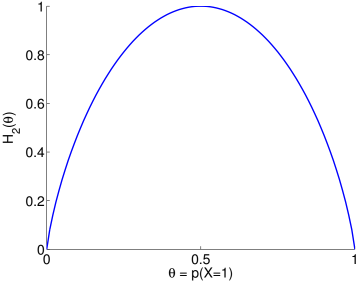

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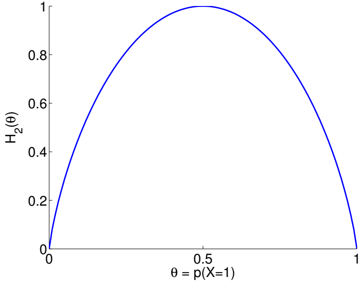

- *Concave* function of the distribution
- Minimum when there is no uncertainty $\theta = 1$ or $\theta = 0$
- Maximum when there is maximum uncertainty $\theta = 0.5$
- For $\theta = 0.5$ (e.g. a fair coin) $H_2(X) = 1$ bit.

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The entropy of this rv is given by:

$$H(X) = -\sum_{i=1}^{32} p(i) \log_2 p(i) = -\sum_{i=1}^{32} \frac{1}{32} \log_2 \frac{1}{32} = \log_2 32 = 5 \text{ bits.}$$

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- This agrees with the number of bits needed to describe X
- In this case all the outcomes have representations of the same length

Categorical distributions with 30 different states:

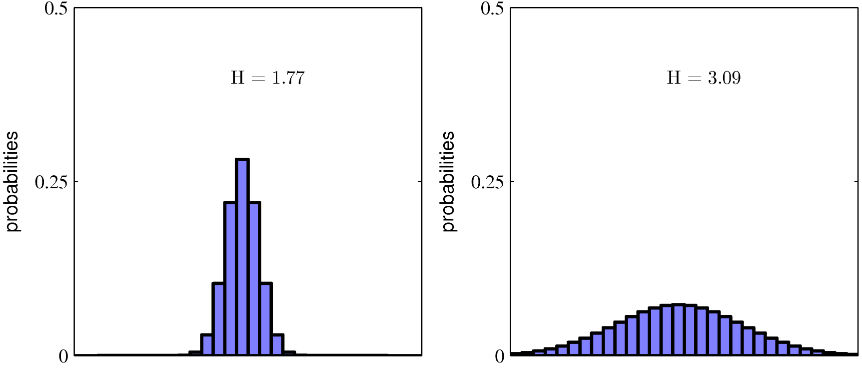

Figure from Bishop, PRML, 2006)

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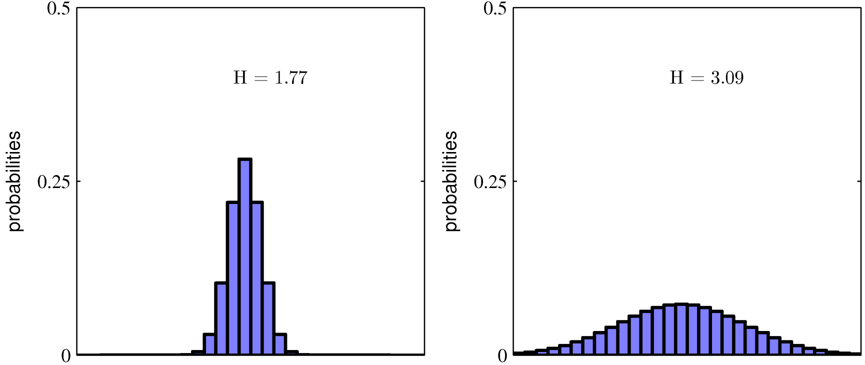

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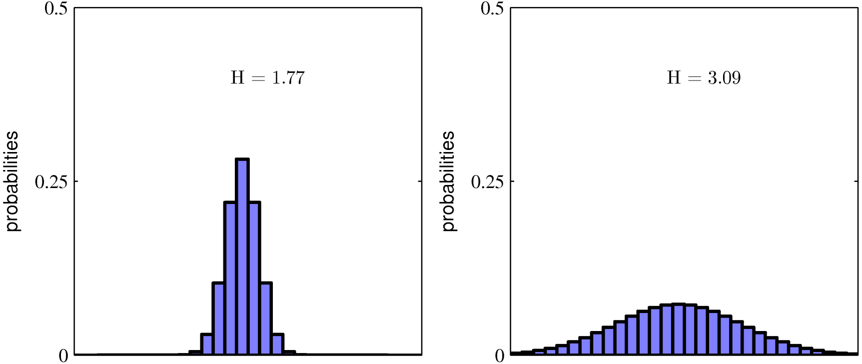

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Entropy of a Random Variable
Example 3 — Categorical Distribution

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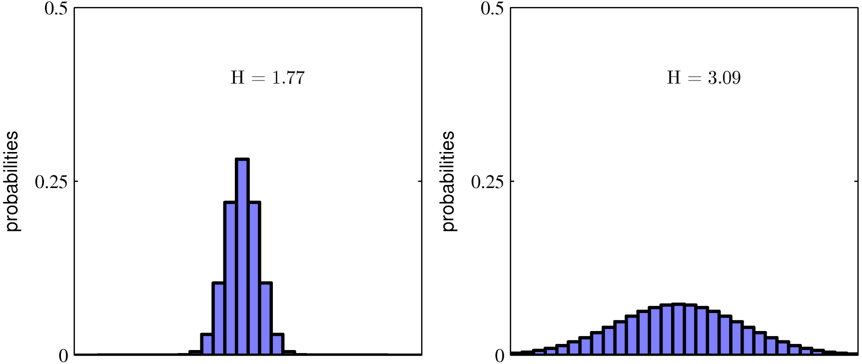

Figure from Bishop, PRML, 2006)

- The more sharply peaked the lower the entropy
- The more evenly spread the higher the entropy
- Maximum for *uniform* distribution: $H(X) = -\log \frac{1}{30} \approx 3.40$ nats
  - When will the entropy be minimum?

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- Let p_i be the probability of each state, with $i = 1, \ldots, |\mathcal{X}|$

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The entropy is maximized if $\mathbf{p}$ is uniform:

$$H(X) \leq \log|\mathcal{X}|$$

With equality iff $p_i = \dfrac{1}{|\mathcal{X}|}$ for all i

Joint Entropy

The joint entropy $H(X, Y)$ of a pair of discrete random variables with joint distribution $p(X, Y)$ is given by:

$$H(X, Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log \frac{1}{p(x, y)}$$
$$= \mathbb{E}_{X,Y} \left[\log \frac{1}{p(X, Y)} \right]$$

Joint Entropy:

Independent Random Variables

if X and Y are statistically independent we have that:

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&= \sum_{x \in \mathcal{X}} p(x) \log \frac{1}{p(x)} + \sum_{y \in \mathcal{Y}} p(y) \log \frac{1}{p(y)} \\
&= H(X) + H(Y)
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Entropy is additive for independent random variables

Conditional Entropy

The conditional entropy of Y given $X = x$ is the entropy of the probability distribution $p(Y|X = x)$:

$$H(Y|X = x) = \sum_{y \in \mathcal{Y}} p(y|X = x) \log \frac{1}{p(y|X = x)}$$

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The conditional entropy of Y given X, is the average over X of the conditional entropy of Y given $X = x$:

$$\begin{aligned} H(Y|X) &= \sum_{x \in \mathcal{X}} p(x) H(Y|X = x) \\ &= \sum_{x \in \mathcal{X}} p(x) \sum_{y \in \mathcal{Y}} p(y|x) \log \frac{1}{p(y|x)} \end{aligned}$$

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This measures the average uncertainty that remains about Y when X is known.

Conditional Entropy — Cont'd

We can re-write the conditional entropy as follows:

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Note the expectation is not wrt the conditional distribution but wrt the joint distribution p(X,Y)

Chain Rule

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The uncertainty of X and Y is the uncertainty of X plus the uncertainty of Y given X

Entropy:

- Measure of uncertainty of a random variable
- Amount of information on average to describe the random variable
- Relation to code-length
- Relation to minimum number of binary questions
- The entropy of a random variable depends on its probability distribution

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We will extend this definition to measure the "distance" between two distributions

- $D(p||q)$: Distance/divergence between two distributions

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- Machine Learning: Approximating a posterior $p(X)$ with $q(X)$

Relative Entropy
Introduction

- $D(p||q)$: Distance/divergence between two distributions
- Machine Learning: Approximating a posterior $p(X)$ with $q(X)$
- Information Theory: Inefficiency of assuming q when the true distribution is p
  - $p(X)$: Can construct a code with average description length $H(p)$
  - $q(X)$: Would need $H(p) + D(p\|q)$ bits on average to describe the r.v.

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A measure of divergence between two distributions is the *relative entropy* or *Kullback-Leibler divergence*

Relative Entropy

Definition

The relative entropy or Kullback-Leibler (KL) divergence between two probability distributions $p(X)$ and $q(X)$ is defined as:

$$\mathrm{KL}(p \| q) = \sum_{x \in \mathcal{X}} p(x) \log \frac{p(x)}{q(x)} = \mathbb{E}_{p(X)} \left[\log \frac{p(X)}{q(X)} \right].$$

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- Note:
  - Both $p(X)$ and $q(X)$ are defined over the same alphabet $\mathcal{X}$
  - KL in statistics
- Conventions:

$$0 \log \frac{0}{0} \stackrel{\text{def}}{=} 0 \qquad 0 \log \frac{0}{q} \stackrel{\text{def}}{=} 0 \qquad p \log \frac{p}{0} \stackrel{\text{def}}{=} \infty$$

Relative Entropy
Properties

- $\mathrm{KL}(p\|q) \geq 0$

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Relative Entropy

Properties

- $\text{KL}(p\|q) \geq 0$
- $\text{KL}(p\|q) = 0 \Leftrightarrow p = q$
- $\text{KL}(p\|q) \neq \text{KL}(q\|p)$

Relative Entropy
Properties

- $\mathrm{KL}(p\|q) \geq 0$
- $\mathrm{KL}(p\|q) = 0 \Leftrightarrow p = q$
- $\mathrm{KL}(p\|q) \neq \mathrm{KL}(q\|p)$
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- $\mathrm{KL}(p\|q) = 0 \Leftrightarrow p = q$
- $\mathrm{KL}(p\|q) \neq \mathrm{KL}(q\|p)$
- Not a true distance since is not symmetric and does not satisfy the triangle inequality
- Very important in machine learning and information theory

Let $X \in \{0, 1\}$ and consider the distributions $p(X)$ and $q(X)$ such that:

$$p(X = 1) = \theta_p \qquad p(X = 0) = 1 - \theta_p$$
$$q(X = 1) = \theta_q \qquad q(X = 0) = 1 - \theta_q$$

What distributions are these?

Relative Entropy

Example (from Cover & Thomas, 2006)

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\end{aligned}
$$

What distributions are these?

Compute $\mathrm{KL}(p\|q)$ and $\mathrm{KL}(q\|p)$ with $\theta_p = \frac{1}{2}$ and $\theta_q = \frac{1}{4}$

$$\mathrm{KL}(p\|q) = \theta_p \log \frac{\theta_p}{\theta_q} + (1-\theta_p)\log\frac{1-\theta_p}{1-\theta_q}$$

Relative Entropy

Example (from Cover & Thomas, 2006) — Cont'd

$$\begin{aligned}
\mathrm{KL}(p\|q) &= \theta_p \log \frac{\theta_p}{\theta_q} + (1-\theta_p) \log \frac{1-\theta_p}{1-\theta_q} \\
&= \frac{1}{2} \log \frac{\frac{1}{2}}{\frac{1}{4}} + \frac{1}{2} \log \frac{\frac{1}{2}}{\frac{3}{4}} = 1 - \frac{1}{2} \log 3 \approx 0.2075 \text{ bits}
\end{aligned}$$

Relative Entropy

Example (from Cover & Thomas, 2006) — Cont'd

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Relative Entropy

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When will $\mathrm{KL}(p\|q) = \mathrm{KL}(q\|p)$?

Mutual Information

Definition

Let X, Y be two r.v. with joint distribution $p(X, Y)$ and marginals $p(X)$ and $p(Y)$:

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Intuitively, how much information, on average, X conveys about Y (or vice versa).

Relationship between Entropy and Mutual Information

We can re-write the definition of mutual information as:

$$I(X;Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}$$

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The average reduction in uncertainty of X due to the knowledge of Y.

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- Since $H(X,Y) = H(X) + H(Y|X)$ we have that:

$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$

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- Since $H(X,Y) = H(X) + H(Y|X)$ we have that:

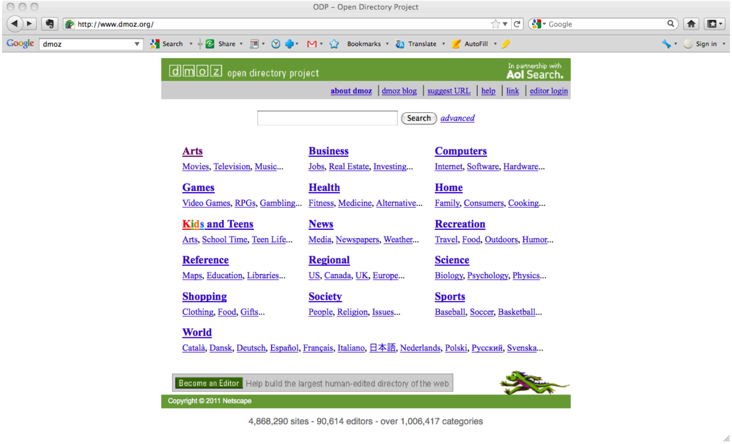
$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$

- Finally:

$$I(X;X) = H(X) - H(X|X) = H(X)$$

Sometimes the entropy is referred to as *self-information*

Recall our problem of document classification:



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Example 2: Feature Selection in Machine Learning (2)

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How can we do this?

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Example 2: Feature Selection in Machine Learning (3)

Greedy selection of features:

1. Start with no features

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- Maximizing $I(X_i; Y)$ wrt X_i is equivalent to minimizing $H(Y|X_i)$

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Potential Problem:

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Potential Problem:

- Optimizing $I(X_i; Y)$ can be seen as maximum relevance
- However, features may be redundant
- What to optimize to avoid including features that are too "similar" to those already included?
  - Maximum relevance - minimum redundancy

Conditional Mutual Information

The conditional mutual information between X and Y given $Z = z_k$:

$$I(X;Y|Z = z_k) = H(X|Z = z_k) - H(X|Y, Z = z_k).$$

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Averaging over Z we obtain:

The conditional mutual information between X and Y given Z:

$$\begin{aligned} I(X; Y|Z) &= H(X|Z) - H(X|Y, Z) \\ &= \mathbb{E}_{p(X,Y,Z)} \log \frac{p(X, Y|Z)}{p(X|Z)p(Y|Z)} \end{aligned}$$

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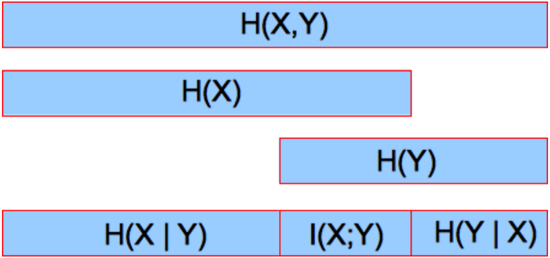
The reduction in the uncertainty of X due to the knowledge of Y when Z is given.

Note that $I(X; Y; Z)$, $I(X|Y; Z)$ are illegal terms while e.g. $I(A, B; C, D|E, F)$ is legal.

Break Down of the Total Entropy



Chain Rule for Mutual Information

Let X, Y, Z be r.v. and recall that:

$$I(X; {\color{blue}Y, Z}) = I({\color{blue}Y, Z}; X) \quad {\color{red}\text{symmetry}}$$

Chain Rule for Mutual Information

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$$\begin{aligned} I(X; Y, Z) &= I(Y, Z; X) \quad \text{symmetry} \\ &= H(Z, Y) - H(Z, Y|X) \quad \text{def. mutual info.} \end{aligned}$$

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$$I(X; Y, Z) = I(X; Y) + I(X; Z|Y) \quad \text{def. mutual info and cond. mutual info}$$

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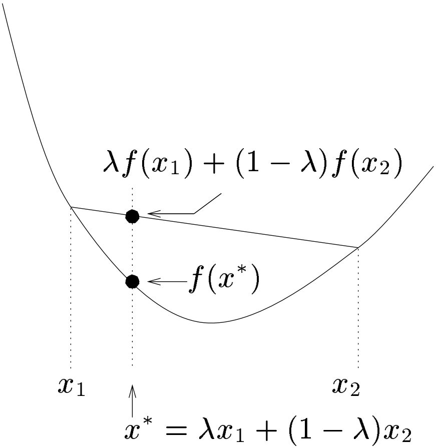
Similarly, by symmetry:

$$I(X; Y, Z) = I(X; Z) + I(X; Y|Z)$$

Convex Functions:

Introduction



$0 \leq \lambda \leq 1$ (Figure from Mackay, 2003)

A function is convex $\smile$ if every cord of the function lies above the function

Definition

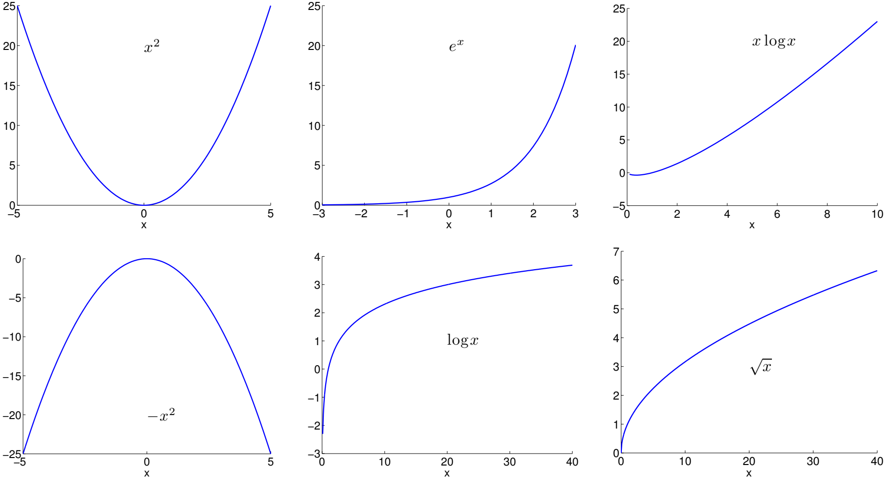
A function $f(x)$ is convex $\smile$ over (a, b) if for all $x_1, x_2 \in (a, b)$ and $0 \le \lambda \le 1$:

$$f(\lambda x_1 + (1 - \lambda)x_2) \le \lambda f(x_1) + (1 - \lambda)f(x_2)$$

We say f is strictly convex $\smile$ if for all $x_1, x_2 \in (a, b)$ the equality holds only for $\lambda = 0$ and $\lambda = 1$.

Similarly, a function f is concave $\frown$ if $-f$ is convex $\smile$, i.e. if every cord of the function lies below the function.

Examples of Convex and Concave Functions



Verifying Convexity

Theorem (Cover & Thomas, Th 2.6.1)

If a function f has a second derivative that is non-negative (positive) over an interval, the function is convex $\smile$(strictly convex $\smile$) over that interval.

This allows us to verify convexity or concavity.

Examples:

- x^2: $\dfrac{d}{dx}\left(\dfrac{d}{dx}(x^2)\right) = \dfrac{d}{dx}(2x) = 2$
- e^x: $\dfrac{d}{dx}\left(\dfrac{d}{dx}(e^x)\right) = \dfrac{d}{dx}(e^x) = e^x$
- $\sqrt{x}$, $x > 0$: $\dfrac{d}{dx}\left(\dfrac{d}{dx}(\sqrt{x})\right) = \dfrac{1}{2}\dfrac{d}{dx}\left(\dfrac{1}{\sqrt{x}}\right) = -\dfrac{1}{4}\dfrac{1}{\sqrt{x^3}}$

Convexity, Concavity and Optimization

if $f(x)$ is concave $\frown$ and there exists a point at which

$$\frac{df}{dx} = 0,$$

then $f(x)$ has a maximum at that point.

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Note: the converse does not hold: if a concave $\frown$ $f(x)$ is maximized at some x, it is not necessarily true that the derivative is zero there.

- $f(x) = -|x|$: is maximized at $x = 0$ where its derivative is undefined
- $f(p) = \log p$ with $0 \leq p \leq 1$, is maximized at $p = 1$ where $\frac{df}{dp} = 1$

Convexity, Concavity and Optimization

if $f(x)$ is concave $\frown$ and there exists a point at which

$$\frac{df}{dx} = 0,$$

then $f(x)$ has a maximum at that point.

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- Generalization to multivariate functions.
- Similarly for convex functions (and minimization)
- Can use second derivative and first derivative together

Jensen's Inequality for Convex Functions

Theorem: Jensen's Inequality

If f is a convex $\smile$ function and X is a random variable then:

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X]).$$

Moreover, if f is strictly convex $\smile$, the equality implies that $X = \mathbb{E}[X]$ with probability 1, i.e X is a constant.

Similarly for a concave $\frown$ function: $\mathbb{E}[f(X)] \leq f(\mathbb{E}[X])$.

Jensen's Inequality Example: The AM-GM Inequality

Recall that for a concave $\frown$ function: $\mathbb{E}[f(X)] \leq f(\mathbb{E}[X])$.

Consider $X \in \{x_1, \ldots, x_N\}$, $X \geq 0$ with uniform probability distribution $\mathbf{p} = (\frac{1}{N}, \ldots, \frac{1}{N})$ and the strictly concave $\frown$ function $f(x) = \log x$:

$$\frac{1}{N} \sum_{i=1}^{N} \log x_i \leq \log \left(\frac{1}{N} \sum_{i=1}^{N} x_i \right)$$

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$$\log\left(\prod_{i=1}^{N} x_i\right)^{\frac{1}{N}} \leq \log\left(\frac{1}{N}\sum_{i=1}^{N} x_i\right)$$

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$$\frac{x_1 + x_2 \ldots + x_N}{N} \geq \sqrt[N]{x_1 x_2 \ldots x_N}$$

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The arithmetic mean is greater than or equal to the geometric mean.

When does the equality hold?

There squares have average area $\overline{A} = 100$ m^2. The average of the lengths of their sides is $\bar{\ell} = 10$ m. What can be said about the largest of the three squares?

Jensen's Inequality

Example (from Mackay, 2003)

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Let $X \in \{\ell_1, \ell_2, \ell_3\}$ denote the length of the side of a square with $\mathbf{p} = (1/3, 1/3, 1/3)$.

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$$\mathbb{E}[X] = 10 \qquad \mathbb{E}[f(X)] = 100,$$

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where $f(x) = x^2$, which is a strictly convex $\smile$ function.

Therefore $f(\mathbb{E}[X]) = \mathbb{E}[f(X)]$, implying that X is a constant and the three lengths $\ell_1 = \ell_2 = \ell_3 = 10$.

Gibbs' Inequality

Theorem

The relative entropy (or KL divergence) between two distributions $p(X)$ and $q(X)$ with $X \in \mathcal{X}$ is non-negative:

$$\mathrm{KL}(p\|q) \geq 0$$

with equality if and only if $p(x) = q(x)$ for all x.

Proof Using Jensen's inequality.

Continuous Random Variables

- Differential Entropy: It is possible to define:

$$H(x) = \mathbb{E}_{p(x)}[-\log p(x)] = -\int p(x) \log p(x) dx$$

However, it does not satisfy all properties, e.g. it can be negative.

- KL Divergence: Similarly, we have seen:

$$\mathrm{KL}(p \| q) \stackrel{\text{def}}{=} \mathbb{E}_{p(x)}\left[\log \frac{p(x)}{q(x)}\right] = \int p(x) \log \frac{p(x)}{q(x)} dx,$$

which always satisfies Gibbs' inequality.

Conditioning Reduces Entropy

Information Cannot Hurt — Example (from Cover & Thomas, 2006)

Let X, Y have the following joint distribution:

	$p(X,Y)$	X = 1	X = 2
Y	1	0	$3/4$
	2	$1/8$	$1/8$

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This is indeed a general statement: $H(X|Y=y_k)$ may be greater than $H(X)$ but the average: $H(X|Y)$ is always less or equal to $H(X)$.

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Information cannot hurt on average.

Conditioning Reduces Entropy

Information Cannot Hurt — Proof

Theorem

$$H(X|Y) \leq H(X),$$

with equality if and only if X and Y are independent.

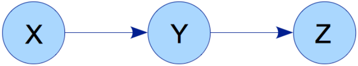
Proof: We simply use the non-negativity of mutual information:

$$\begin{aligned} I(X;Y) &\geq 0 \\ H(X) - H(X|Y) &\geq 0 \\ H(X|Y) &\leq H(X) \end{aligned}$$

with equality if and only if $P(X, Y) = P(X)P(Y)$, i.e X and Y are independent.

Data are helpful, they don't decrease uncertainty on average.

Markov Chain

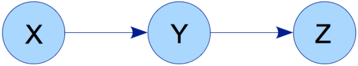


Definition

Random variables X, Y, Z are said to form a Markov chain in that order (denoted by $X \to Y \to Z$) if their joint probability distribution can be written as:

$$p(X, Y, Z) = p(X)p(Y|X)p(Z|Y)$$

Markov Chain



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Consequences:

- $X \to Y \to Z$ if and only if X and Z are conditionally independent given Y. Proof?
- $X \to Y \to Z$ implies that $Z \to Y \to X$.
- If $Z = f(Y)$, then $X \to Y \to Z$

Theorem

$$\text{if } X \to Y \to Z \text{ then: } I(X;Y) \geq I(X;Z)$$

- X is the state of the world, Y is the data gathered and Z is the processed data
- No "clever" manipulation of the data can improve the inferences that can be made from the data
- No processing of Y, deterministic or random, can increase the information that Y contains about X

We have seen that the chain rule for mutual information states that:

$$\begin{aligned} I(X;Y,Z) &= I(X;Y) + I(X;Z|Y) \\ &= I(X;Z) + I(X;Y|Z) \end{aligned}$$

Therefore:

Data-Processing Inequality

Proof

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$$I(X; Y) + \underbrace{I(X; Z|Y)}_{0} = I(X; Z) + I(X; Y|Z)$$ Markov chain assumption

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$$
\begin{aligned}
I(X;Y) + \underbrace{I(X;Z|Y)}_{0} &= I(X;Z) + I(X;Y|Z) \quad \text{Markov chain assumption} \\
I(X;Y) &= I(X;Z) + I(X;Y|Z) \quad \text{but } I(X;Y|Z) > 0
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Corollary

In particular, if $Z = g(Y)$ we have that:

$$I(X;Y) \geq I(X;g(Y))$$

Proof: $X \rightarrow Y \rightarrow g(Y)$ forms a Markov chain.

Functions of the data Y cannot increase the information about X

Corollary

If $X \to Y \to Z$ then $I(X;Y|Z) \leq I(X;Y)$

Proof: We use again the chain rule for mutual information:

$$\begin{aligned} I(X;Y,Z) &= I(X;Y) + I(X;Z|Y) \\ &= I(X;Z) + I(X;Y|Z) \end{aligned}$$

Therefore:

$$I(X;Y) + \underbrace{I(X;Z|Y)}_{0} = I(X;Z) + I(X;Y|Z) \quad \text{Markov chain assumption}$$

$$I(X;Y|Z) = I(X;Y) - I(X;Z) \quad \text{but } I(X;Z) > 0$$

$$I(X;Y|Z) \leq I(X;Y)$$

The dependence between X and Y cannot be increased by the observation of a "downstream" variable.

Summary & Conclusions

- Entropy as a measure of information content
- Computation of entropy of discrete random variables
- Joint and conditional entropies
- Relative entropy
- Convex Functions, Jensen's inequality, Gibbs' inequality
- Reading:
  * Mackay (ITILA, 2003): Sec. 1.2 – 1.5, 2.5, 2.6 – 2.10, 8.1
  * Bishop (PRML, 2006): Sec. 1.6
  * Murphy (MLaPP, 2012): Sec. 2.8